

Splunk Enterprise Portfolio

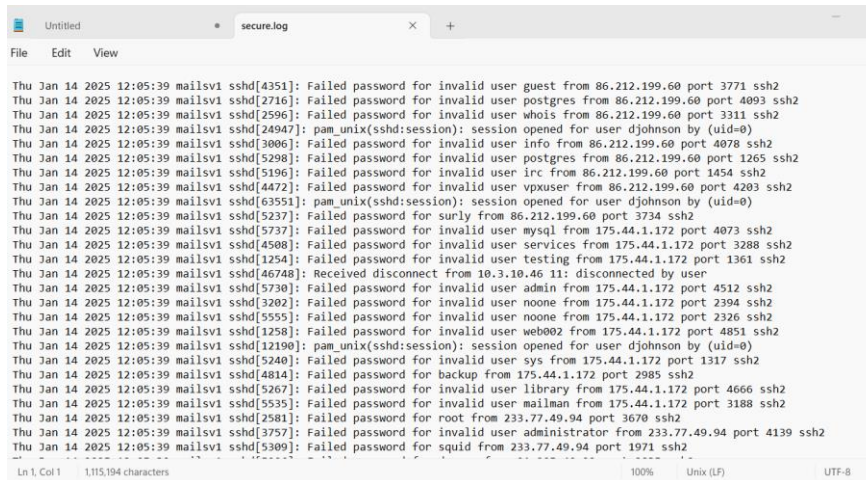
This portfolio showcases my hands-on experience with Splunk through real-world exercises.

I want to demonstrate my ability to analyze security logs, detect threats, create dashboards, and automate alerts.

EXERCISE 1

Note: The data which I used was downloaded from [here](#) we have the following details:

- ✓ **Source:** secure.log
- ✓ **Sourcetype:** emilia.logs
- ✓ **Host:** splunk exercises



```
Thu Jan 14 2025 12:05:39 mailsvl sshd[4351]: Failed password for invalid user guest from 86.212.199.60 port 3771 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[2716]: Failed password for invalid user postgres from 86.212.199.60 port 4093 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[2596]: Failed password for invalid user whois from 86.212.199.60 port 3311 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[24947]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsvl sshd[3006]: Failed password for invalid user info from 86.212.199.60 port 4078 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5298]: Failed password for invalid user postgres from 86.212.199.60 port 1265 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5196]: Failed password for invalid user irc from 86.212.199.60 port 1454 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[4472]: Failed password for invalid user vpxuser from 86.212.199.60 port 4203 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[63551]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsvl sshd[5237]: Failed password for surly from 86.212.199.60 port 3734 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5737]: Failed password for invalid user mysql from 175.44.1.172 port 4073 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[4508]: Failed password for invalid user services from 175.44.1.172 port 3288 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[1254]: Failed password for invalid user testing from 175.44.1.172 port 1361 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[46748]: Received disconnect from 10.3.10.46 11: disconnected by user
Thu Jan 14 2025 12:05:39 mailsvl sshd[5730]: Failed password for invalid user admin from 175.44.1.172 port 4512 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[3202]: Failed password for invalid user noone from 175.44.1.172 port 2394 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5555]: Failed password for invalid user noone from 175.44.1.172 port 2326 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[1258]: Failed password for invalid user web002 from 175.44.1.172 port 4851 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[12190]: pam_unix(sshd:session): session opened for user djohnson by (uid=0)
Thu Jan 14 2025 12:05:39 mailsvl sshd[5240]: Failed password for invalid user sys from 175.44.1.172 port 1317 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[4814]: Failed password for backup from 175.44.1.172 port 2985 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5267]: Failed password for invalid user library from 175.44.1.172 port 4666 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5535]: Failed password for invalid user mailman from 175.44.1.172 port 3188 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[2581]: Failed password for root from 233.77.49.94 port 3670 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[3757]: Failed password for invalid user administrator from 233.77.49.94 port 4139 ssh2
Thu Jan 14 2025 12:05:39 mailsvl sshd[5309]: Failed password for squid from 233.77.49.94 port 1971 ssh2
```

New Search

index=main source="secure.log" sourcetype="emilia.logs"

✓ 9,829 events (before 1/28/25 12:30:13.000 PM) No Event Sampling ▼

Events (9,829) Patterns Statistics Visualization

✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect



✓ Format ▼ Show: 50 Per Page ▼ View: List ▼

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS # date_hour 1 # date_minute 1 # date_month 1 # date_second 2 # date_wday 7 # date_year 1 # date_zone 1 # index 1 # linecount 1 # punct 9		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[5276]: Failed password for invalid user appserver from 194.8.74.23 port 3351 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1039]: Failed password for root from 194.8.74.23 port 3768 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[5258]: Failed password for invalid user testuser from 194.8.74.23 port 3626 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[21881]: pam_unix(sshd:session): session closed for user nsharpie by (uid=0) host = splunk exercises source = secure.log sourcetype = emilia.logs
INTERESTING FIELDS # date_hour 1 # date_minute 1 # date_month 1 # date_second 2 # date_wday 7 # date_year 1 # date_zone 1 # index 1 # linecount 1 # punct 9		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1165]: Failed password for apache from 194.8.74.23 port 4604 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3760]: Failed password for invalid user mongodb from 194.8.74.23 port 2472 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[4998]: Failed password for mail from 194.8.74.23 port 1552 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1930]: Failed password for games from 194.8.74.23 port 3007 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs

1) Finding failed logins:

New Search

Save As ▼ Create Table View Close

index=main source="secure.log" sourcetype="emilia.logs" "Failed password"

All time ▼ 🔍

✓ 8,154 events (before 1/28/25 12:31:32.000 PM) No Event Sampling ▼

Job ▼ || || → 📎 ⬇️ 📄 Smart Mode ▼

Events (8,154) Patterns Statistics Visualization

✓ Timeline format ▼ — Zoom Out + Zoom to Selection × Deselect

1 day per column



✓ Format ▼ Show: 50 Per Page ▼ View: List ▼

< Prev 1 2 3 4 5 6 7 8 ... Next >

< Hide Fields	≡ All Fields	i	Time	Event
SELECTED FIELDS # date_hour 1 # date_minute 1 # date_month 1 # date_second 2 # date_wday 5 # date_year 1 # date_zone 1 # index 1 # linecount 1 # punct 3 # splunk_server 1 # timeendpos 1 # timestartpos 1		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[5276]: Failed password for invalid user appserver from 194.8.74.23 port 3351 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1039]: Failed password for root from 194.8.74.23 port 3768 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[5258]: Failed password for invalid user testuser from 194.8.74.23 port 3626 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1165]: Failed password for apache from 194.8.74.23 port 4604 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
INTERESTING FIELDS # date_hour 1 # date_minute 1 # date_month 1 # date_second 2 # date_wday 5 # date_year 1 # date_zone 1 # index 1 # linecount 1 # punct 3 # splunk_server 1 # timeendpos 1 # timestartpos 1		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3760]: Failed password for invalid user mongodb from 194.8.74.23 port 2472 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[4998]: Failed password for mail from 194.8.74.23 port 1552 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[1930]: Failed password for games from 194.8.74.23 port 3007 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[5801]: Failed password for invalid user desktop from 194.8.74.23 port 2285 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
+ Extract New Fields		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3759]: Failed password for nagios from 194.8.74.23 port 3769 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3759]: Failed password for nagios from 194.8.74.23 port 3769 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3759]: Failed password for nagios from 194.8.74.23 port 3769 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs
		>	1/21/25 12:05:40.000 PM	Thu Jan 21 2025 12:05:40 mailsvl sshd[3759]: Failed password for nagios from 194.8.74.23 port 3769 ssh2 host = splunk exercises source = secure.log sourcetype = emilia.logs

It came out that we have about 8154 failed passwords, which is a lot!

New Search

index=main source="secure.log" sourcetype="emilia.logs" "Failed password" | stats count

✓ 8,154 events (before 1/28/25 12:34:11.000 PM) No Event Sampling ▼

Events Patterns **Statistics (1)** Visualization

Show: 20 Per Page ▼ / Format ▼ ☒ Preview: On

count ↕

8154

2) Finding the 10 most frequent failed password attempts

- Initiated in order to help us identifying patterns or anomalies in the log
- The conclusion would be that several attempts are listed under users like *operator*, *administrator*, *nginx*, *web002*, which make us think that these could be brute-force attacks, where attackers try many different user accounts.
- On the other hand, the facts that all the failed attempts are happening at the same time, could indicate that an automated tool was being used to guess passwords.
- Also, the failed attempts were caused by a variety of source IP addresses, suggesting that these are likely remote access attempts (SSH).

New Search

```
index=main source="secure.log" sourcetype="emilia.logs" "failed password"
| stats count by _raw
| sort - count
| head 10
```

✓ 8,154 events (before 1/28/25 12:46:40.000 PM) No Event Sampling ▼

Events Patterns **Statistics (10)** Visualization

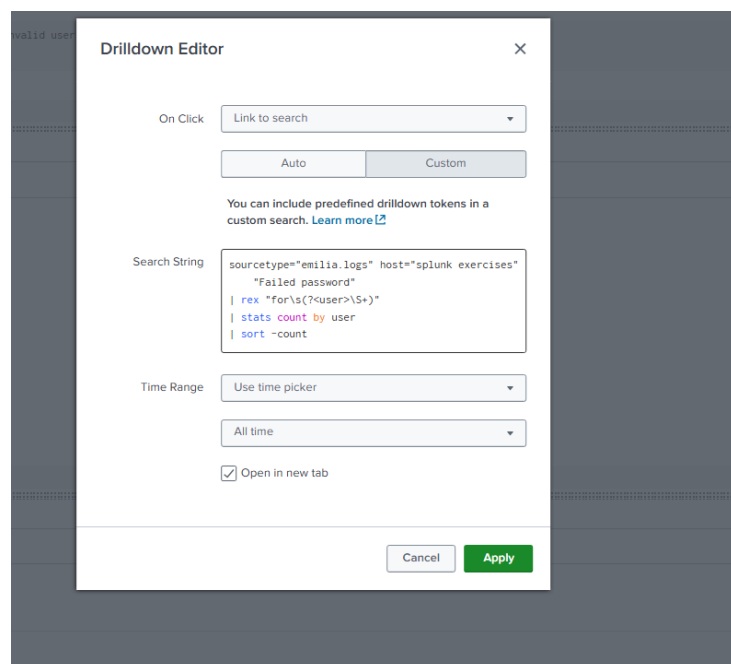
Show: 20 Per Page ▼ Format  Preview: On

_raw ↕

```
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1004]: Failed password for invalid user operator from 194.8.74.23 port 2382 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1019]: Failed password for beyonce from 121.254.179.199 port 3776 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1025]: Failed password for nagios from 201.42.223.29 port 4430 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1035]: Failed password for invalid user administrator from 128.241.220.82 port 1967 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1036]: Failed password for invalid user operator from 128.241.220.82 port 1640 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1037]: Failed password for invalid user web002 from 97.117.230.183 port 4123 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1040]: Failed password for squid from 195.2.240.99 port 1112 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1053]: Failed password for root from 211.140.3.183 port 3504 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1055]: Failed password for invalid user nginx from 194.215.205.19 port 3061 ssh2
Fri Jan 15 2025 12:05:39 mailsv1 sshd[1059]: Failed password for gopher from 121.254.179.199 port 3173 ssh2
```

3) Creating the dashboard with the top users with failed logins

I've used the following query



Drilldown Editor

On Click: Link to search ▼

Auto Custom

You can include predefined drilldown tokens in a custom search. [Learn more](#)

Search String:

```
sourcetype="emilia.logs" host="splunk exercises"
"Failed password"
| rex "for\s(?<user>\S+)"
| stats count by user
| sort -count
```

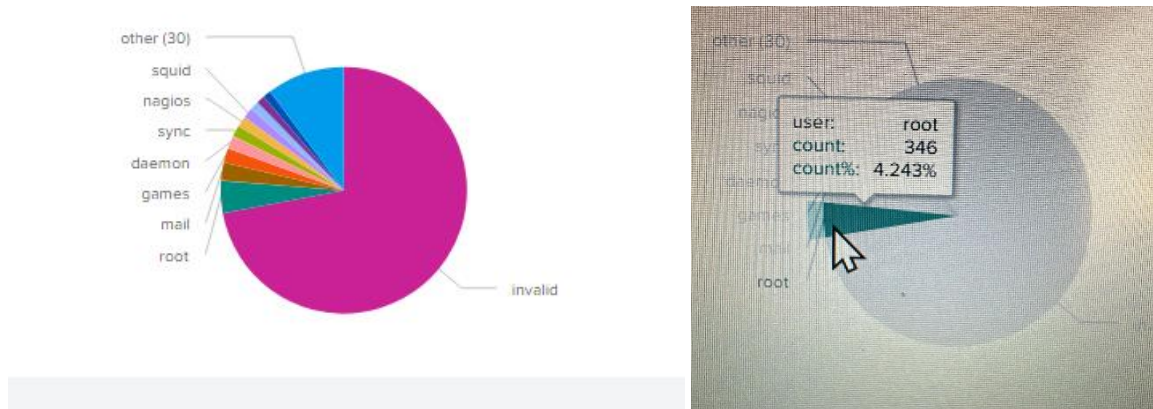
Time Range: Use time picker ▼

All time ▼

☒ Open in new tab

Cancel Apply

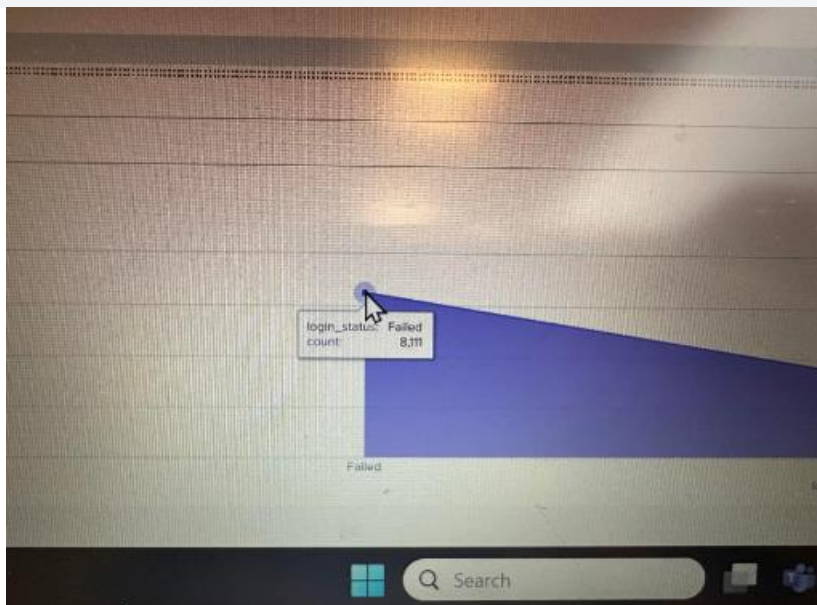
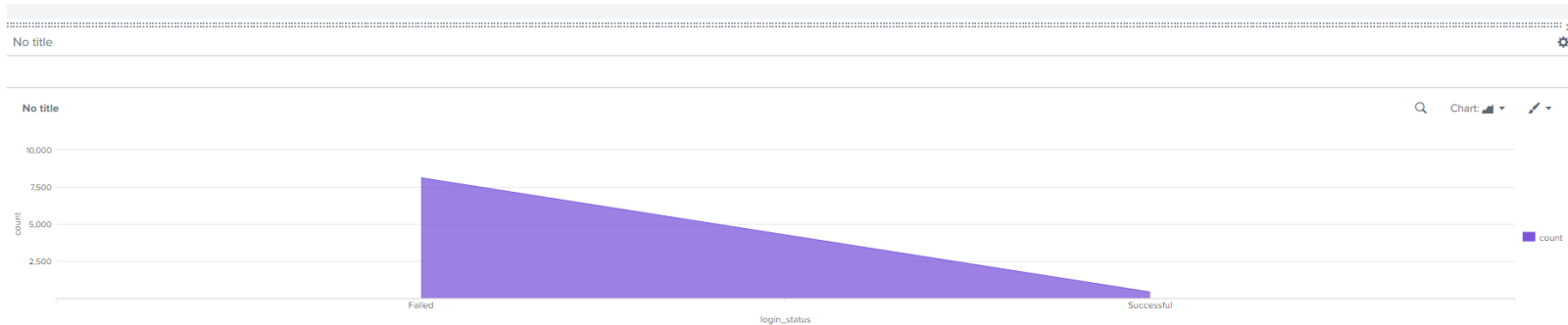
And I've obtained the following chart



4) Failed versus successful logins

I've used the following string in order to view on the chart the difference between failed and successful logins.

```
sourcetype="emilia.logs" host="splunk exercises" ("Failed password" OR "Accepted password")  
| eval login_status=if(match(_raw, "Failed password"), "Failed", "Successful")  
| stats count by login_status
```



The number of failed logins was 8.111 while the successful logins were 423. (*as per the report created and used)

Advices:

I would recommend to review who has access to SSH and to harden the authentication methods; instead of passwords we could use SSH keys or other MFA.

The other advice would be to block certain suspicious IP addresses, which are blocked or flagged in firewall logs.

And last but not the least, to set up alerts in Splunk for repeated failed login attempts, thing which I already did (as shown below).

Save As Alert [X]

Settings

Title: Failed Login Alert

Description: Triggers when an IP fails to log in more than 5 times

Permissions: Private | Shared in App

Alert type: Scheduled | Real-time

Run on Cron Schedule ▼

Time Range: All time ▶

Cron Expression: */5 * * * *

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires: 24 | hour(s) ▼

Trigger Conditions

Trigger alert when: Number of Results ▼

is greater than ▼ | 0

Trigger: Once | For each result

Throttle ? ☒

Suppress triggering for: 1 | hour(s) ▼

Trigger Actions

+ Add Actions ▼

Cancel Save

Since we suspect a brute force attack, setting the interval to every 5 minutes could be a good option.

Also, I checked the option "Throttle" in order to mute alerts for the same IP for 1 hour to prevent alert spam.

Save As Alert



Throttle ? ☒

Suppress triggering for

1

hour(s) ▼

Trigger Actions

+ Add Actions ▼

When triggered



✉ Send email

Remove

To

changeme@example.com



Comma separated list of email addresses.
Email addresses represented by tokens are
validated only at the time of the search.
[Show CC and BCC](#)

Priority

Normal ▼

Subject

Splunk Alert: \$name\$

The email subject, recipients and message
can include tokens that insert text based on
the results of the search. [Learn More](#)

Message

The alert condition for '\$name\$'
was triggered.



Include

- | | |
|---|---|
| ☒ Link to Alert | ☒ Link to Results |
| ☐ Search String | ☐ Inline Table ▼ |
| ☐ Trigger Condition | ☐ Attach CSV |

Cancel

Save

EXERCISE 2

Note: The data used for analysis was taken from [here](#) (*Static information about Zeus binaries*) and uploaded in Splunk under these details:

- ✓ **Source/file name:** zeus.zip
- ✓ **Sourcetype:** json-too_small
- ✓ **Host:** testnow

We will perform a malware analysis.

- 1) First we will check when the events occurred and where they came from; we've noticed that they all have the same host.

New Search			Save As ▾	Create Table View	Close
index=testdata sourcetype=json-too_small			All time ▾ 		
table _time host source					
✓ 8,010 events (before 1/28/25 3:54:57:000 PM) No Event Sampling ▾			Job ▾ ▢ ↗ ⚙ ⬇ 🗨 Verbose Mode ▾		
Events (8,010) Patterns Statistics (8,010) Visualization					
Show: 20 Per Page ▾ Format ▾  Preview: On			< Prev 1 2 3 4 5 6 7 8 ... Next >		
_time ▴	host ▴	source ▴			
2016-09-13 06:20:53	testnow	zeus.zip:.\zeus/zeusbin_fffd8b831f289b5f5764d952624442.ex0.json			
2016-09-13 06:20:52	testnow	zeus.zip:.\zeus/zeusbin_ffe2186f1379afb58e5ae49f35cc8a10.ex0.json			
2016-09-13 06:20:52	testnow	zeus.zip:.\zeus/zeusbin_ffd7d4c60f005a403dba135529601ed1.ex0.json			
2016-09-13 06:20:52	testnow	zeus.zip:.\zeus/zeusbin_ffd5389d2f062940b58163d3b81fdb02.ex0.json			
2016-09-13 06:20:52	testnow	zeus.zip:.\zeus/zeusbin_ffce4579b59eb7e3e5577519be7eacb0.ex0.json			
2016-09-13 06:20:51	testnow	zeus.zip:.\zeus/zeusbin_ffcaf8a2f2f59e0f7b165d085842bd17.ex0.json			
2016-09-13 06:20:50	testnow	zeus.zip:.\zeus/zeusbin_ffc3fa4a7dd1d3872e990afbe0036d00.ex0.json			
2016-09-13 06:20:49	testnow	zeus.zip:.\zeus/zeusbin_ffbaf01e391e7c5f2cf846d7f84812e5.ex0.json			
2016-09-13 06:20:48	testnow	zeus.zip:.\zeus/zeusbin_ffba31a88f3920149417a78b52fbf6a3.ex0.json			
2016-09-13 06:20:48	testnow	zeus.zip:.\zeus/zeusbin_ffb1f7802b2ea174e0c9b72565b06450.ex0.json			
2016-09-13 06:20:48	testnow	zeus.zip:.\zeus/zeusbin_ffa8bccc9732db6d47e50f7715d10562.ex0.json			
2016-09-13 06:20:47	testnow	zeus.zip:.\zeus/zeusbin_ff88d64553b23da7580805c4d0984e4fc.ex0.json			
2016-09-13 06:20:47	testnow	zeus.zip:.\zeus/zeusbin_ff751120570b57c53d5985f8baaf0e52.ex0.json			
2016-09-13 06:20:46	testnow	zeus.zip:.\zeus/zeusbin_ff589d252f226770445cb428196914e4.ex0.json			
2016-09-13 06:20:46	testnow	zeus.zip:.\zeus/zeusbin_ff503cd80e9b8a58645cd15b0bfb3df4.ex0.json			
2016-09-13 06:20:44	testnow	zeus.zip:.\zeus/zeusbin_ff3dbe4fb645f12e28143c8da77c0e0a.ex0.json			
2016-09-13 06:20:44	testnow	zeus.zip:.\zeus/zeusbin_ff09a848374a6435eb4433883cb86b4.ex0.json			
2016-09-13 06:20:43	testnow	zeus.zip:.\zeus/zeusbin_ff07394036050ce7b1a987dc5e77c570.ex0.json			
2016-09-13 06:20:43	testnow	zeus.zip:.\zeus/zeusbin_ffef06eed3fda791d177ba1f2dec2096.ex0.json			

2) I've found no "attack", hence 2207 events with "error".

The "error parsing" could indicate possible file corruption; one more indication of this would be *IMAGE_SCN_MEM_WRITE* and *IMAGE_SCN_MEM_EXECUTE*, which could lead to obfuscation.

New Search

Save As>Create Table ViewClose

index=testdata sourcetype=json-too_small "error"

All time

2,207 events (before 1/28/25 4:13:18.000 PM)No Event Sampling

Job

Timeline formatZoom OutZoom to SelectionDeselect

1 month per column

FormatShow: 50 Per PageView: List

12345678Next

< Hide FieldsAll Fields

SELECTED FIELDS

a host 1

a source 100+

a sourcetype 1

INTERESTING FIELDS

a Directories[]Size.FileOffset 61

a Directories[]Size.Offset 1

a Directories[]Size.Value 100+

a Directories[]Structure 16

a Directories[]VirtualAddress.FileOffset 61

a Directories[]VirtualAddress.Offset 1

a Directories[]VirtualAddress.Value 100+

a DllCharacteristics[] 5

DOS_HEADER.e.cbip.FileOffset 1

DOS_HEADER.e.cbip.Offset 1

DOS_HEADER.e.cbip.Value 4

DOS_HEADER.e.cp.FileOffset 1

DOS_HEADER.e.cp.Offset 1

DOS_HEADER.e.cp.Value 4

DOS_HEADER.e.cparhdr.FileOffset 1

DOS_HEADER.e.cparhdr.Offset 1

DOS_HEADER.e.cparhdr.Value 2

i

Time

Event

> 9/13/16 6:20:53.000 AM

{ "DllCharacteristics": { "IMAGE_DLLCHARACTERISTICS_TERMINAL_SERVER_AWARE", "IMAGE_DLLCHARACTERISTICS_NX_COMPAT" }, "FileHeader": { "NumberOfSections": { "FileOffset": 230, "Value": 4, "Offset": 2 }, "TimeStamp": { "FileOffset": 232, "Value": "0x508bFEa6 [Sat Oct 27 15:32:54 2012 UTC]", "Offset": 4 }, "PointerToSymbolTable": { "FileOffset": 236, "Value": 0, "Offset": 8 }, "NumberOfSymbols": { "FileOffset": 240, "Value": 0, "Offset": 12 }, "Machine": { "FileOffset": 228, "Value": 332, "Offset": 8 }, "Characteristics": { "FileOffset": 246, "Value": 259, "Offset": 18 }, "SizeOfOptionalHeader": { "FileOffset": 244, "Value": 224, "Offset": 16 }, "Structure": "IMAGE_FILE_HEADER", "ParsingWarnings": ["Error parsing section 0. SizeOfRawData is larger than file.", "Error parsing section 1. SizeOfRawData is larger than file.", "Error parsing section 1. PointerToRawData points beyond the end of the file.", "Error parsing section 2. SizeOfRawData is larger than file.", "Error parsing section 2. PointerToRawData points beyond the end of the file.", "Error parsing section 3. SizeOfRawData is larger than file.", "Error parsing section 3. PointerToRawData points beyond the end of the file.", "Possibly corrupt file. AddressOfEntryPoint lies outside the file. AddressOfEntryPoint: 0x306a2", "Corrupt header \\\"IMAGE_IMPORT_DESCRIPTOR\\\" at file offset 333692. Exception: 'Data length less than expected header length.'", "Corrupt header \\\"IMAGE_RESOURCE_DIRECTORY\\\" at file offset 409600. Exception: 'Data length less than expected header length.'", "Invalid resources directory. Can't parse directory data at RVA: 0x90000", "Corrupt header \\\"IMAGE_DEBUG_DIRECTORY\\\" at file offset 280544. Exception: 'Data length less than expected header length.'", "Corrupt header \\\"IMAGE_LOAD_CONFIG_DIRECTORY\\\" at file offset 331552. Exception: 'Data length less than expected header length.'", "Directories: [{ \"VirtualAddress\": { \"FileOffset\": 344, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_EXPORT\", \"Size\": { \"FileOffset\": 348, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 352, \"Value\": 340348, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_IMPORT\", \"Size\": { \"FileOffset\": 356, \"Value\": 80, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 360, \"Value\": 589824, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_RESOURCE\", \"Size\": { \"FileOffset\": 364, \"Value\": 35256, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 368, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_SECURITY\", \"Size\": { \"FileOffset\": 380, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 384, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_BASERELOC\", \"Size\": { \"FileOffset\": 388, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 392, \"Value\": 287200, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_DEBUG\", \"Size\": { \"FileOffset\": 396, \"Value\": 28, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 400, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_COPYRIGHT\", \"Size\": { \"FileOffset\": 404, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 408, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_GLOBALPTR\", \"Size\": { \"FileOffset\": 412, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 416, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_ILS\", \"Size\": { \"FileOffset\": 420, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 424, \"Value\": 338208, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_LOAD_CONFIG\", \"Size\": { \"FileOffset\": 428, \"Value\": 64, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 432, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_BOUND_IMPORT\", \"Size\": { \"FileOffset\": 436, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 440, \"Value\": 286720, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_IAT\", \"Size\": { \"FileOffset\": 444, \"Value\": 424, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 448, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_DELAY_IMPORT\", \"Size\": { \"FileOffset\": 452, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 456, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_COM_DESCRIPTOR\", \"Size\": { \"FileOffset\": 460, \"Value\": 0, \"Offset\": 4 }, { \"VirtualAddress\": { \"FileOffset\": 464, \"Value\": 0, \"Offset\": 0 }, \"Structure\": \"IMAGE_DIRECTORY_ENTRY_RESERVED\", \"Size\": { \"FileOffset\": 468, \"Value\": 0, \"Offset\": 4 }], \"OPTIONAL_HEADER\": { \"DllCharacteristics\": { \"FileOffset\": 318, \"Value\": 33024, \"Offset\": 70 }, \"BaseOfCode\": { \"FileOffset\": 268, \"Value\": 4096, \"Offset\": 20 }, \"MajorLinkerVersion\": { \"FileOffset\": 250, \"Value\": 12, \"Offset\": 2 }, \"MinorOperatingSystemVersion\": { \"FileOffset\": 290, \"Value\": 1, \"Offset\": 42 }, \"MinorLinkerVersion\": { \"FileOffset\": 251, \"Value\": 0, \"Offset\": 3 }, \"MajorSubsystemVersion\": { \"FileOffset\": 296, \"Value\": 5, \"Offset\": 48 }, \"MajorOperatingSystemVersion\": { \"FileOffset\": 288, \"Value\": 5, \"Offset\": 40 }, \"SizeOfStackCommit\": { \"FileOffset\": 324, \"Value\": 4096, \"Offset\": 76 }, \"SizeOfUninitializedData\": { \"FileOffset\": 260, \"Value\": 0, \"Offset\": 12 }, \"NumberOfRvaAndSizes\": { \"FileOffset\": 340, \"Value\": 16, \"Offset\": 92 }, \"SizeOfHeapReserve\": { \"FileOffset\": 328, \"Value\": 1048576, \"Offset\": 80 }, \"LoaderFlags\": { \"FileOffset\": 336, \"Value\": 0, \"Offset\": 88 }, \"SizeOfHeapCommit\": { \"FileOffset\": 332, \"Value\": 4096, \"Offset\": 84 }, \"SizeOfStackReserve\": { \"FileOffset\": 320, \"Value\": 1048576, \"Offset\": 72 }, \"Structure\": \"IMAGE_OPTIONAL_HEADER\", \"SizeOfHeaders\": { \"FileOffset\": 380, \"Value\": 1024, \"Offset\": 60 }, \"Subsystem\": { \"FileOffset\": 316, \"Value\": 2, \"Offset\": 68 }, \"Magic\": { \"FileOffset\": 248, \"Value\": 267, \"Offset\": 0 }, \"MinorImageVersion\": { \"FileOffset\": 234, \"Value\": 0, \"Offset\": 46 }, \"MajorImageVersion\": { \"FileOffset\": 292, \"Value\": 0, \"Offset\": 44 }, \"Value\": 2, \"Offset\": 68 }, \"Magic\": { \"FileOffset\": 248, \"Value\": 267, \"Offset\": 0 }, \"MinorImageVersion\": { \"FileOffset\": 234, \"Value\": 0, \"Offset\": 46 }, \"MajorImageVersion\": { \"FileOffset\": 292, \"Value\": 0, \"Offset\": 44 }, \"Value\": 2, \"Offset\": 68 } }

Recommendations:

- to implement File Integrity Monitoring, to monitor system files or unexpected file access patterns

-to implement EDR against suspicious memory-based activity due to *IMAGE_SCN_MEM_WRITE* and *IMAGE_SCN_MEM_EXECUTE* flags

- to use if it's the case sandboxing tools for testing suspicious files or activities

- to use application whitelisting so that only trusted, pre-approved applications are allowed to run